DESIGN OF MACHINE ELEMENTS

II B.TECH II Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6ME20	PCC	3	0	0	3	40	60	100

COURSE OVERVIEW :

This course deals the systematic approach to design, standardization, design and manufacturing considerations, engineering materials properties and their selection, simple and compound stresses in machine elements, theories of failure under static and dynamic loading situations, Design of riveted joints, welded joints, bolted joints, keys and cotter joints, shafts, shaft couplings and springs.

COURSE OUTCOMES:

The course should enable the students to

1. Explain various types of design, various considerations for design and theories of failure.
2. Design riveted, welded and bolted joints.
3. Design keys, cotter and knuckle joints.
4. Design shafts and shaft couplings.
5. Design Mechanical Springs.

UNIT-I INTRODUCTION & FATIGUE

Introduction: General considerations & Manufacturing considerations for design of machine elements, Mechanical properties of materials, and preferred numbers.
Simple Stresses, Simple bending, Pure Torsion, Combined bending and torsion, Static theories of failure.
Fatigue: Types of Fatigue loading, S-N Curve – Endurance Limit, Factors effecting Endurance Limit, Stress Concentration - Theoretical stress concentration factor - Fatigue stress concentration factor - Notch sensitivity- Design for fluctuating stresses-endurance limit-Goodman line, Soderberg line and Gerber's Parabola.

UNIT-II DESIGN OF RIVTED, WELDED & BOLTED JOINTS

Riveted Joints: Cold riveting and hot riveting, types of riveted joints, terminology, methods of failure of riveted joints-strength equations-efficiency of riveted joints-eccentrically loaded riveted joints
Welded Joints: Types, Advantages, Dis-advantages, Design of fillet welds-axial loads-circular fillet welds subjected to torsion and bending.
Design of bolts: Design of bolts with pre-stresses, bolts of uniform strength, bolted joints under eccentric loading.

UNIT-III DESIGN OF KEYS, COTTER& KNUCKLE JOINTS

Keys, Cotter& Knuckle Joints: Design of key-stresses in keys-cotter joints-spigot &socket, sleeve & cotter, Gib & cotter joints & Knuckle Joints.

UNIT-IV DESIGN OF SHAFTS & SHAFT COUPLINGS

Design of shafts: Design of solid & hollow shafts for strength & rigidity- Design of shafts for complex loads-shaft sizes-Design of shaft for Belt drives.
Shaft couplings: Rigid couplings-Muff, split muff and flange couplings-Flexible couplings-PIN Bush couplings

UNIT-V DESIGN OF MECHANICAL SPRINGS

Mechanical Springs: Stresses & Deflection of helical springs-extension of compression springs- springs for static & fatigue loading of helical springs- energy storage capacity-helical torsion springs, Laminated Springs
Text Books:
1. V. Bandari (2011), *A Text Book of Design of Machine Elements*, 3rd edition, Tata McGraw hill education (P) ltd, New Delhi, India. 2. R. L. Norton (2006), *Machine Design (An Integrated approach*), 2nd edition, Pearson Publishers, Chennai, India.
Reference Books:
1. Shigley, J.E, (2011), *Mechanical Engineering Design*, 9th Edition, Tata McGraw-Hill, New Delhi, India. 2. Dr.P.C Sharma, D.K Agarwal, Machine Design, S.K Kataria& Sons